

CS604 Software Project Management Comprehensive Notes

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Unit 1: Software Project Planning & Estimation

COCOMO (Constructive Cost Model) - Boehm:

Basic COCOMO Formula:

- Effort (E) = $a * (KLOC)^b$ [Person-Months]
- Development Time (D) = $c * (E)^d$ [Months]
- Persons Required (P) = E / D .

Project Modes:

1. Organic (Small teams, familiar environment): $a = 2.4$, $b = 1.05$.
2. Semidetached (Medium teams, mixed experience): $a = 3.0$, $b = 1.12$.
3. Embedded (Rigid constraints, hardware coupling): $a = 3.6$, $b = 1.20$.

Unit 2: PERT & CPM Scheduling

CPM (Critical Path Method):

Deterministic activity durations. The Critical Path is the longest sequence of dependent activities from project start to finish.

Float / Slack = Late Start (LS) - Early Start (ES) = Late Finish (LF) - Early Finish (EF). Activities on the critical path have Zero Float.

PERT (Program Evaluation and Review Technique):

Probabilistic durations with 3 time estimates:

- Expected Time $T_e = (t_o + 4*t_m + t_p) / 6$
where t_o is optimistic time, t_m is most likely time, and t_p is pessimistic time.
- Variance $\sigma^2 = ((t_p - t_o) / 6)^2$.

Unit 3: Risk Management & Agile Methodologies

Risk Management Cycle:

1. Risk Identification: Brainstorming, checklists, taxonomy.
2. Risk Analysis: Probability x Impact = Risk Exposure.
3. Risk Mitigation, Monitoring, and Management (RMMM Plan).

Agile Scrum Framework:

- Roles: Product Owner, Scrum Master, Development Team.
- Ceremonies: Sprint Planning, Daily Standup (15 min), Sprint Review, Sprint Retrospective.
- Artifacts: Product Backlog, Sprint Backlog, Burndown Chart.

